

Software Requirements Specification (SRS) Document

Brief problem statement

The project proposes to build an efficient, speedy and accurate pipeline that can build context based on user-filled conversations to fill forms for a defined purpose. Size of the model, latency of the application and accuracy of the filled fields are of crucial focus. Support for image and audio input may be considered as extensions, if time permits.

The system enables users to select or create domain-specific forms and automatically fill them using conversational text. Primary benefits of the project include reduction in time spent for manual form-filling, live structured data extraction and dynamic form evolution for real-life scenarios. This product aligns with Product Labs' goal of building AI-assisted productivity pipelines optimized for performance and deployability.

System Requirements

The Chat-Based Form Filler is a lightweight web application designed for efficient AI-based form extraction with low latency and scalable deployment.

Hardware Requirements

- **Server:** Minimum 4 GB RAM (8 GB recommended), dual-core CPU (Optional GPU for faster inference). Currently planning on deploying on Product Labs server with sufficient CPU and GPU resources.
- **Client:** Any modern device with a web browser. Microphone required for audio input (R2)

Software Requirements

Operating Environment

- Docker-based deployment using a lightweight Linux runtime (`python:3.11-slim`)

Backend

- FastAPI (Python 3.11)
- PyTorch, Hugging Face Transformers
- Models: FunctionGemma, DistilBERT (subject to change as we progress, keeping overall architecture the same)

Frontend

- HTML5, CSS3, JavaScript

Database

- MongoDB (MongoDB Atlas for cloud hosting)

Performance Requirements

- Target extraction latency is not a rigid factor, always prefer accuracy of output over latency (as directed by the client). For typical text inputs, anything less than two minutes is acceptable.
- The above goes for real-time updates in live mode as well.
- Supports multiple concurrent users.

Security & Deployment

- Role-based access control
- Containerized deployment enabling scalability and cloud/on-premise hosting

Scalability

- The architecture must support increasing users, forms, and conversations.
- New domains and form templates should be added without redesigning the system.

Accuracy & Quality

- Extracted form fields should maintain high accuracy with clear confidence indicators (if available).
- The system must allow manual correction to ensure final data reliability.

Users profile

The system will be used by administrators, data entry personnel, and employees across different organizations, as the application is domain-agnostic.

Administrators manage the system configuration. They create and modify form templates, define fields, and they are typically comfortable with technology and require flexible configuration, access control, and system reliability.

Data entry personnel use the system intensively to fill large volumes of forms. Speed, accuracy, and minimal manual effort are critical for them. They are likely to rely on real-time text or audio input, so the interface must be fast, clear, and easy to correct.

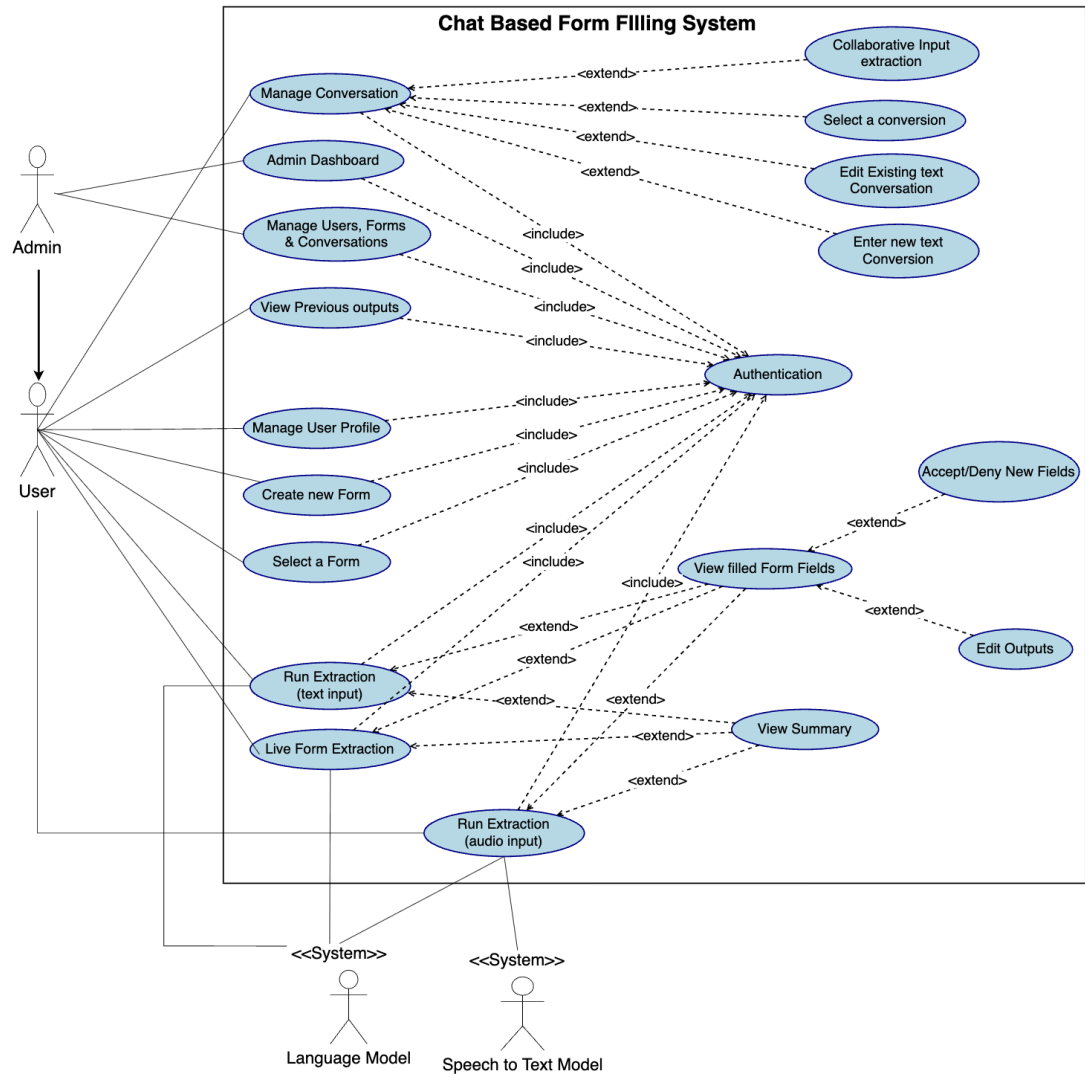
Employees/end users provide information for forms as part of their regular work. Their technical skills may vary, and they may prefer voice input. The system must be simple, intuitive, and allow pausing and resuming form completion.

Feature requirements (described using use cases)

No.	User Case Name	Description	Release
	Create new Form	A user can create a new form to fit the domain of the conversation they would wish to run an extraction on. They can customize the fields and corresponding description.	R1

	Run Extraction (text input)	A user can run the model pipeline to receive form-filled fields on a text conversation typed into the designated section.	R1
	View filled Form Fields	A user can view the input conversation alongside the filled fields to draw parallels.	R1
	Manage User Profile	A user maintains a profile for all conversations and extractions performed on the system.	R2
	View Previous outputs	A user can view previous extraction results. A version control system is implemented, where the user may toggle between different versions.	R1
	Select a Form	A user can choose a form among the existing ones for extraction.	R1
	Edit Outputs	A user can edit and correct the filled and missing fields produced by the model pipeline.	R1
	Accept/Deny New Fields	A user can approve/deny the new fields suggested by the model pipeline and edit the values as well.	R1
	Admin Dashboard	The admin can view all the users and their usage history.	R2
	Run Extraction (audio input)	A user can run the model pipeline to receive form-filled fields on an audio conversation recorded at or attached to the designated section.	R2
	Collaborative Input extraction	Different user profiles can collaborate to complete a text conversation and run extractions simultaneously.	R2
	View Summary	Every extraction will contain a summary for user review along with form-filled values.	R1
	Manage Users, Forms and Conversations	The admin can view all registered users, saved conversations, and saved forms and delete/edit any of them if required.	R2
	Authentication	All users are to be authenticated for privacy upon launching the website/app.	R2
15.	Edit Existing text Conversation	A user can edit an existing conversation by adding or removing messages and run the extraction again for newer outputs.	R1
16.	Select a conversation	A user can select already existing default conversations and run the extraction	R1

Use case diagram



Use case description

Use Case Number:	UC-01
Use Case Name:	Create new Form
Overview:	A user can create a new form to fit the domain of the conversation they would wish to run an extraction on. They can customize the fields and corresponding description.
Actors:	User
Pre condition:	1. The User must successfully login to the system after suitable authentication.

Flow:	Main Flow: 1. The User can click on the "+ Add New Form" button present on their dashboard. 2. The User can add a title and description to the form for future use. 3. The User can add relevant fields along with their data types (e.g., String, Int). 4. The User can click on "Submit" to successfully create the form.
	Alternate Flows: 4. The User clicks on "Cancel" to not create the form.
Post Condition:	The system generates a unique Form ID, saves the form, and the user may choose the form created to run an extraction.

Use Case Number:	UC-02
Use Case Name:	Run Extraction (text input)
Overview:	A user can run the model pipeline to receive form-filled fields on a text conversation typed into the designated section.
Actors:	User
Pre condition:	1. The User must successfully login to the system after suitable authentication. 2. The User must either select or create a form to run the extraction on.
Flow:	Main Flow: 1. The User can dynamically add speakers to the conversation using the "Add Speaker" input. 2. The User can select the speaker by clicking on the required speaker name iteratively while adding messages and dialogues. 3. The User can click on "Save & Extract" to run the model pipeline for the outputs.
	Alternate Flows: 3. The User clicks on "Back" and leaves the page. 3. The User clicks on "Save" and does not run an extraction.
Post Condition:	The User can view a summary, the outputs and edit/add context after completion.

Use Case Number:	UC-03
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Use Case Name:	View filled Form Fields
Overview:	A user can view the input conversation alongside the filled fields to draw parallels.
Actors:	User
Pre condition:	1. The User must successfully login. 2. The User must select/create a form. 3. The User must type out a conversation and run the extraction.
Flow:	Main Flow: 1. The User can view the form fields filled with values according to the model pipeline. 2. The view includes both the input conversation and the filled values. 3. The User may choose to view the AI summary of the conversation.
	Alternate Flows: 3. The User chooses not to view the outputs.
Post Condition:	The User may choose to view the summary, edit fields, add context to the conversation based on output satisfaction.

Use Case Number:	UC-04
Use Case Name:	Manage User Profile
Overview:	A user maintains a profile for all conversations and extractions performed on the system.
Actors:	User
Pre condition:	1. The User must successfully login to the system after suitable authentication.
Flow:	Main Flow: 1. The User can view registration details - email, password, username etc. 2. The User can change the email, password and username etc. All these steps will require sufficient verification of account ownership beforehand.
	Alternate Flows: 2. User authentication or verification fails.
Post Condition:	The User can maintain persistent and secure access to their information and conversations.

Use Case Number:	UC-05
Use Case Name:	View Previous outputs
Overview:	A user can view previous extraction results. A version control system is implemented, where the user may toggle between different versions.
Actors:	User
Pre condition:	1. The User must successfully login to the system after suitable authentication. 2. The User must have run extraction on at least one conversation before availing this use-case for effect.
Flow:	Main Flow: 1. In the "View Past Outputs" or "View Logs" section, the User can view the previous runs executed from the account. 2. The User can view both - the input conversation and the filled fields along with summary and other metadata.
	Alternate Flows: 2. The User clicks on "Back" to exit.
Post Condition:	The User can choose to edit/add context to older conversations. Version history is maintained for reliability.

Use Case Number:	UC-06
Use Case Name:	Select a Form
Overview:	A user can choose a form among the existing ones for extraction.
Actors:	User
Pre condition:	1. The User must successfully login to the system after suitable authentication.
Flow:	Main Flow: 1. On the dashboard, the User can view a list of all available forms. 2. The user can select the form of choice by clicking on its ID or Name to begin entering a conversation.
Post Condition:	The User can select, type or record a conversation for extraction.

Use Case Number:	UC-07
Use Case Name:	Edit Outputs
Overview:	A user can edit and correct the filled and missing fields produced by the model pipeline by modifying the conversation context.
Actors:	User
Pre condition:	1. The User must successfully login. 2. The User must have run an extraction.
Flow:	Main Flow: 1. The User can click the "Edit Conversation" button on the outputs screen. 2. The User can modify the raw transcript (adding missing details or correcting errors). 3. The User clicks "Update & Re-Extract" to submit the new context.
Post Condition:	The system saves the new conversation version and outputs the corrected, updated form fields.

Use Case Number:	UC-08
Use Case Name:	Approve/Deny New Fields
Overview:	A user can approve/deny the new fields suggested by the model pipeline and edit the values as well.
Actors:	User
Pre condition:	1. The User must have run an extraction on a conversation.
Flow:	Main Flow: 1. The User can review the suggested form fields and currently filled values given by the model pipeline. 2. If the User deems the fields relevant, they may choose to approve the field. 3. The User may edit the output for a recommended field. 4. The User may also reject the field on the grounds of irrelevance.
	Alternate Flows: 4. The User may choose not to approve/deny the fields - the fields are defaulted to deny.
Post Condition:	The User can submit the form with desired fields and values.

Use Case Number:	UC-09
Use Case Name:	Admin Dashboard
Overview:	The admin can view all the users and their usage history.
Actors:	Admin
Pre condition:	1. The Admin must successfully login to the system after suitable authentication.
Flow:	Main Flow: 1. The Admin navigates to the Admin Dashboard. 2. The Admin can access the list of all registered users through the Admin dashboard. 3. The Admin clicks on a user profile to view more details about them, the conversations they entered and their form extraction history.
Post Condition:	The admin can successfully view and monitor all user activity on the platform.

Use Case Number:	UC-10
Use Case Name:	Run Extraction (audio input)
Overview:	A user can run the model pipeline to receive form-filled fields on an audio conversation recorded at or attached to the designated section.
Actors:	User
Pre condition:	1. The User must select or create a form to run the extraction on.
Flow:	Main Flow: 1. The User can choose the number of speakers participating in the conversation. 2. The User can either record audio or attach a recording in the input section of the screen. 3. The User can click on "Run Extraction" for the outputs. 4. The User can review the outputs similar to the ones for text conversations.
Post Condition:	The User can view a summary, the outputs and edit/add context after completion.

Use Case Number:	UC-11
Use Case Name:	Collaborative Input extraction
Overview:	Different user profiles can collaborate to complete a text conversation and run extractions simultaneously.
Actors:	User
Pre condition:	1. The Users must successfully login to the system after suitable authentication. 2. The Users must select the same form to run the extraction on. Both the users must have permissions to participate in the conversation.
Flow:	Main Flow: (Flow to be determined in Release 2).
Post Condition:	Both users can view the seamlessly updated conversation context and corresponding extraction results.

Use Case Number:	UC-12
Use Case Name:	View Summary
Overview:	Every extraction will contain a summary for user review along with form-filled values.
Actors:	User
Pre condition:	1. The User must type out a conversation and run the extraction on it.
Flow:	Main Flow: 1. On the “View Outputs” page, the User may choose to opt for the “View Summary” button. 2. The summary includes all relevant information gathered from the attached conversation.
Post Condition:	The User can validate the summary and re-run an extraction if needed.

Use Case Number:	UC-13
Use Case Name:	Manage Users, Forms and Conversations

Overview:	The admin can view all registered users, saved conversations, and saved forms and delete/edit any of them if required.
Actors:	Admin
Pre condition:	1. The Admin must successfully login to the system after suitable authentication.
Flow:	Main Flow: 1. The Admin can view all registered users, saved conversations, saved forms and form extractions. 2. The Admin can select one of the users, conversations, forms or form extractions to expand and view more information about them. 3. The admin can delete/edit the selected item.
	Alternate Flows: 3. The Admin can go back to the page containing all registered users, saved conversations, saved forms and form extractions.
Post Condition:	The delete/edit updates for the selected item will be reflected in the page displaying all the items to the Admin and any process involving these items will use the updated items(if they still exist).

Use Case Number:	UC-14
Use Case Name:	Authentication
Overview:	All users are to be authenticated for privacy upon launching the website/app.
Actors:	User, Admin
Pre condition:	The User and Admin have access to the domain.
Flow:	Main Flow: 1. The correct login-signup details are entered. 2. All credentials are verified.
	Alternate Flows: 1. The login credentials are incorrect. 2. The user is not authenticated and access to the website is not provided.
Post Condition:	The User/Admin is securely logged into their respective dashboard.

Use Case Number:	UC-15
Use Case Name:	Edit Existing text Conversation
Overview:	A user can edit an existing conversation by adding or removing messages and run the extraction again for newer outputs.
Actors:	User
Pre condition:	1. The User must be able to select an existing previous conversation linked to the chosen form.
Flow:	Main Flow: 1. The User can choose the "Edit / Add Context" button on the page of conversation of choice. 2. The User may add bubbles with speakers and dialogues of choice. 3. The User can opt to "Update & Re-Extract" on the current conversation state.
	Alternate Flows: 2. The user can choose not to make any edit and go back.
Post Condition:	The User can view the summary, filled form fields or error messages upon failure.

Use Case Number:	UC-16
Use Case Name:	Select a conversation
Overview:	A user can select already existing default conversations and run the extraction
Actors:	User
Pre condition:	1. The User has selected a form from the domain of choice.
Flow:	Main Flow: 1. The User can view existing conversations with detail (Conversation ID, timestamps, etc.). 2. The User can select a conversation and opt to "Run Extraction on Latest Version" or "Edit/Add context".
	Alternate Flows: 2. The user can select a new conversation, read it and go back.
Post Condition:	The User can view the summary, filled form fields or error messages upon failure.